MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive Text Sources

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ICLR 2026

Paper

Funded by the European Union

TL;DR

We present **MixtureVitae**, a 422B-token open-access pretraining corpus built from permissive-first sources that closes the performance gap to leading non-permissive datasets, proving that legally robust data can train capable LLMs.

- **Permissive-first, risk-mitigated recipe:** Three-tier licensing scheme (CC-BY, Apache, public domain + government works) with shard-level provenance metadata
- **Single-stage pretraining with instruction & reasoning data:** Front-loads synthetic instruction, math, and code data typically reserved for post-training
- **1.7B base model matches instruction-tuned baselines:** Outperforms SmolLM2-1.7B-Instruct on GSM8K, HumanEval, and MBPP using 36x fewer tokens (300B vs ~11T)

Data Curation Pipeline

Positive inclusion strategy: Instead of retroactive filtering of broad web scrapes, we positively select sources based on auditable permissive status.

- **Permissiveness filtering:** Allowlist of governmental domains + permissive license keyword search, excluding restrictive terms
- **Safety filtering:** CSAM & offensive content removal via keyword blocklists; targeted filtering of biographies of living persons
- **Quality filtering:** Removal of base64-encoded text and duplicative headers/footers
- **Deduplication:** Local intra-dataset prefix-based exact matching only; intentionally no global fuzzy dedup to preserve stylistic and domain diversity (following FineWeb-Edu findings)
- **Domain-aware mixing:** Documents clustered by base URL, sentences concatenated within domain for coherent training examples

Data Sources

- **Web-scale:** Nemotron-CC, MGACorpus, FineFineWeb (incl. synthetic rephrased web text)
- **Curated:** SEC EDGAR, MegaWika, TxT360, arXiv, peS2o, PubMed, The Stack v1, USPTO, EuroPat, DM-Math, VALID, YouTube Commons, Open License Corpus
- **Instruction:** Magpie Collection, UltraFeedback, NVIDIA SFT blend, P3 (few-shot & MC format)
- **Reasoning:** Glaive-AI, OpenThoughts, CaseHOLD, OpenScience, OpenManus-RL
- **Math & Code:** MetaMathQA, OpenMathInstruct-2, DART-MATH, Nemo-Math, Prism-Math, LingCoder, StarCoder

Dataset Composition

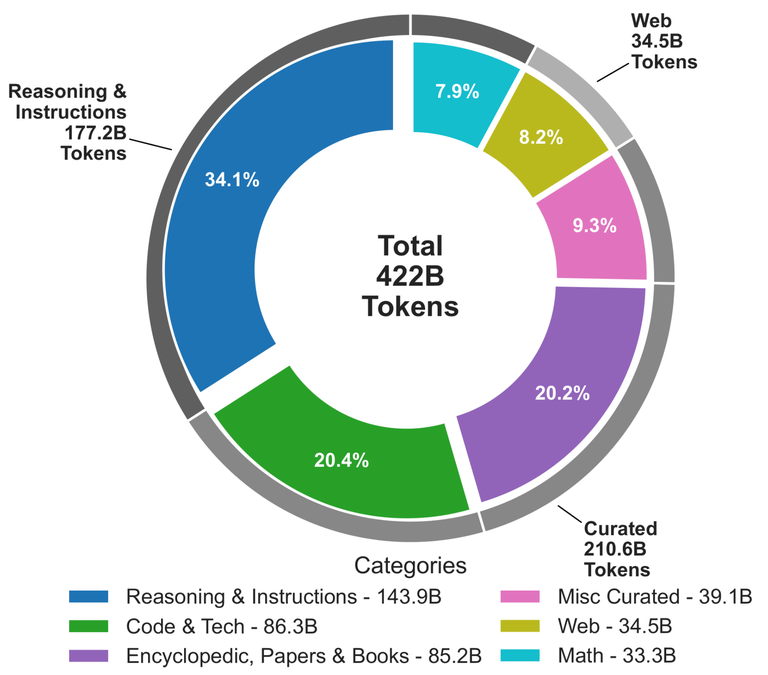

Figure 1. MixtureVitae composition by top-level category and content domain. The 422B-token corpus spans web data, curated sources, and instruction/reasoning datasets.

License & Provenance Distribution

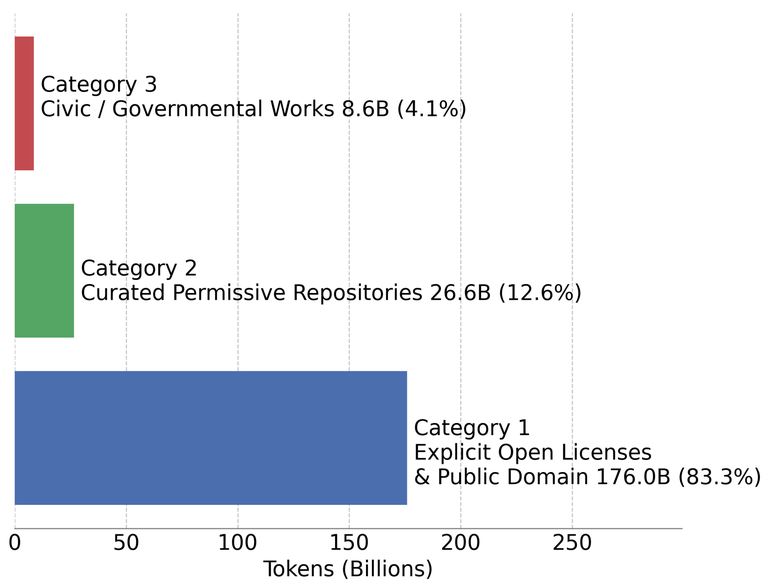

Figure 2. Legal provenance tiers of MixtureVitae. All sources fall into permissive-first or risk-mitigated tiers with shard-level annotations.

Synthetic & Instruction Data

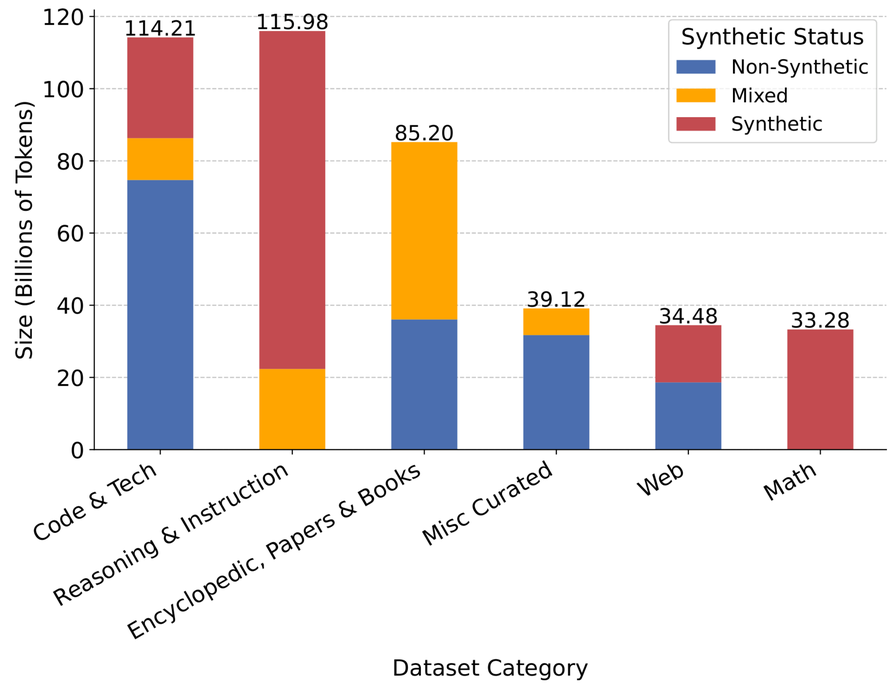

Figure 3. Composition by data origin. Bars show non-synthetic (human-written), mixed, and fully synthetic proportions per domain.

Overall Performance (300B Tokens)

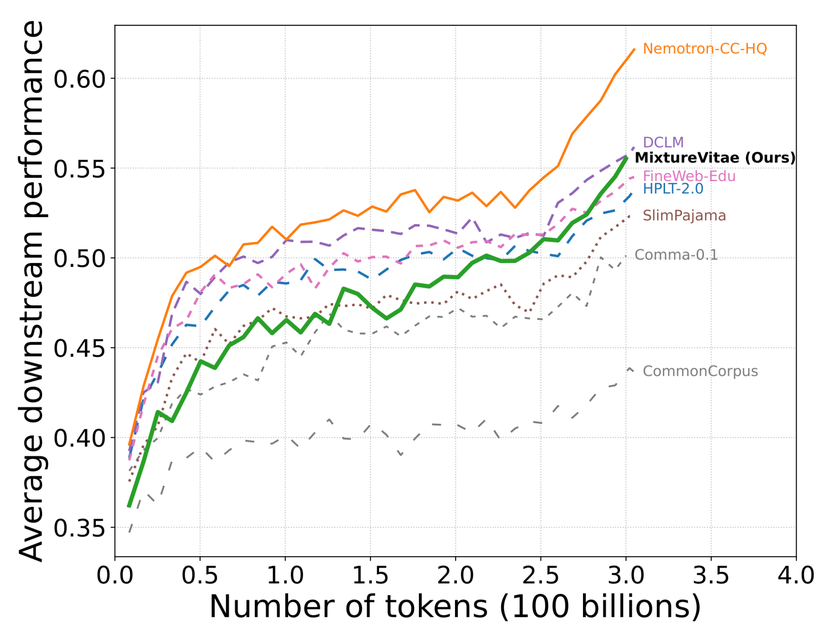

Figure 4. Average performance across 10 benchmarks for 1.7B models. MixtureVitae outperforms all permissive baselines and approaches DCLM and FineWeb-Edu at 300B tokens.

MMLU Performance

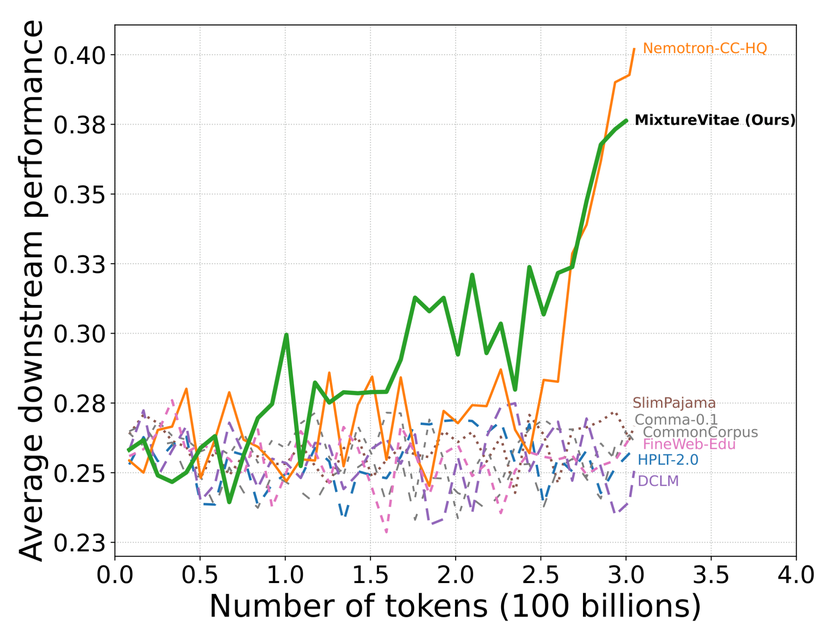

Figure 5. Performance on MMLU. MixtureVitae significantly outperforms all baselines; only Nemotron-CC-HQ catches up at ~260B tokens.

Ablation Study

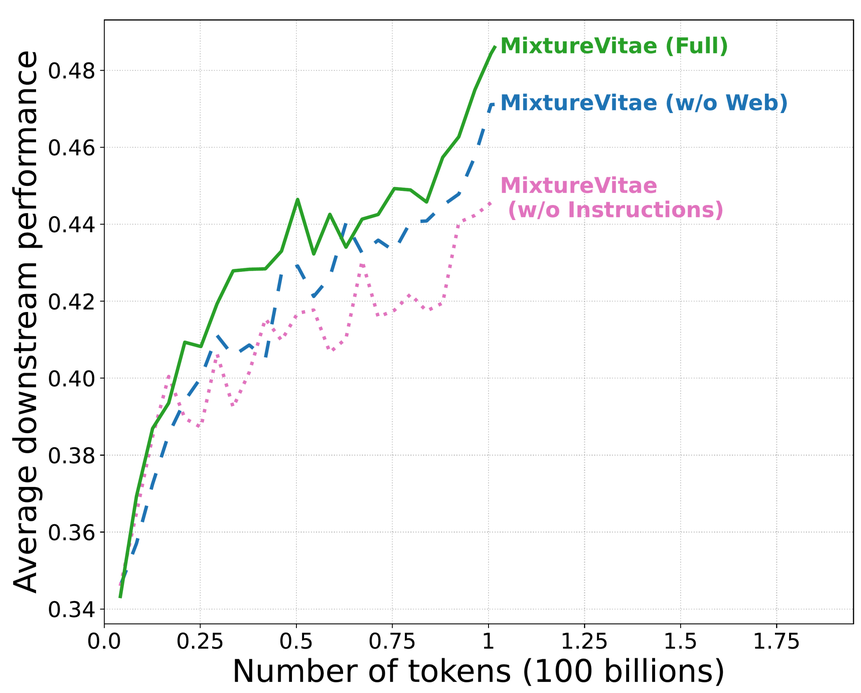

Figure 7. Ablation on MixtureVitae components (1.3B model, 100B tokens). Removing Instructions causes the largest drop, confirming instruction and reasoning data as the critical performance driver.

Math, Code & Instruction Results

1.7B MixtureVitae base model **outperforms SmolLM2-1.7B-Instruct** on GSM8K, HumanEval, and MBPP, despite using 36x fewer tokens.

Dataset	Tokens	IF-Eval	GSM8K	HumanEval	MBPP	Avg
MixtureVitae	**300B**	**0.19**	**0.53**	**0.32**	**0.38**	**0.36**
Comma-0.1	300B	0.19	0.06	0.13	0.22	0.15
CommonCorpus	300B	0.13	0.02	0.05	0.05	0.06
C4	300B	0.20	0.02	0.00	0.00	0.06
SlimPajama	300B	0.14	0.02	0.05	0.00	0.05
DCLM	300B	0.13	0.02	0.01	0.01	0.04
Nemotron-CC-HQ	300B	0.09	0.03	0.02	0.00	0.03
Trained for 1T tokens						
FineWeb-Edu	1T	0.20	0.03	0.00	0.00	0.06
DCLM	1T	0.15	0.03	0.00	0.01	0.05
Other models (~11T tokens)						
SmolLM2-1.7B	11T	0.18	0.31	0.01	0.35	0.21
SmolLM2-1.7B-Instruct	11T	**0.28**	0.37	0.28	0.37	0.33

Three-Tier Licensing Scheme

Tier 1 — Explicit Open Licenses & Public Domain
CC0, CC-BY, Apache 2.0, MIT, BSD, public domain. Synthetic data from permissive models & seeds.

Tier 2 — Curated Permissive with Upstream Opacity
(a) Permissive corpora with partial provenance (The Stack V1, Wikipedia). (b) Synthetic data with non-permissive generators (~4% of corpus, isolated for transparency).

Tier 3 — Civic / Governmental Works
US federal works (statutory public domain), government websites, regulatory notices with strong public-purpose rationale for reuse.

Decontamination Analysis

MixtureVitae's strong performance is **not an artifact of benchmark leakage**. A fully decontaminated model performs slightly *better* than the original.

- **13-gram overlap scan** across all 345M documents; contamination rate negligible for most benchmarks (≤0.01%)
- **Original vs. decontaminated test sets:** Performance consistent (e.g., GSM8K 0.53 → 0.54, MBPP 0.38 → 0.38)
- **Full corpus decontamination:** Retrained 1.7B model after removing all flagged documents; results *improved*

Original vs. Decontaminated Scores

Dataset	GSM8K	GSM8K-D	MBPP	MBPP-D	IFEval	IFEval-D
MixtureVitae	**0.53**	**0.54**	**0.38**	**0.38**	**0.19**	**0.23**
SmolLM2	0.30	0.30	0.35	0.35	0.17	0.20
Comma-0.1	0.06	0.06	0.21	0.23	0.18	0.20
DCLM	0.01	0.02	0.01	0.00	0.12	0.13
FineWeb	0.02	0.01	0.00	0.00	0.18	0.20
Nemotron-CC-HQ	0.03	0.02	0.00	0.00	0.09	0.10

License Distribution

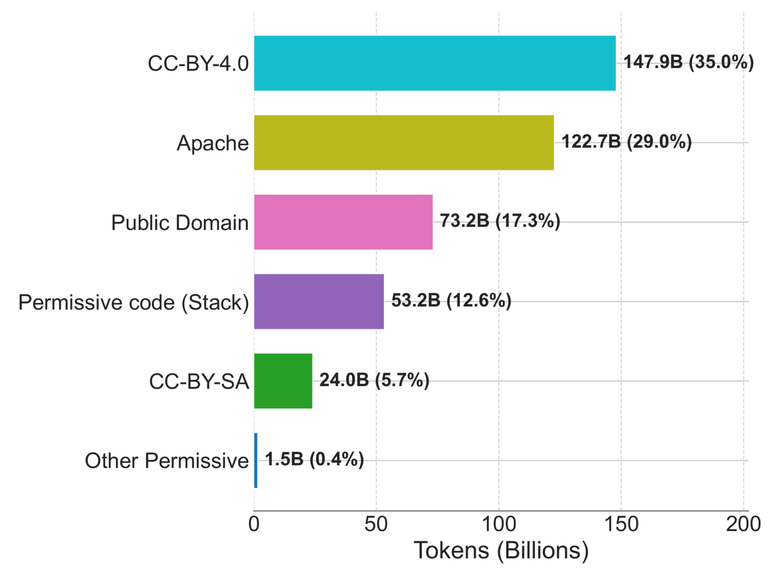

Figure 6. Token distribution by governing license. The corpus is composed entirely of data under permissive licenses or public domain.

Conclusion & Key Takeaways

MixtureVitae demonstrates that **permissive-first data with high instruction and reasoning density** can provide a practical and risk-mitigated foundation for training capable LLMs.

- **Shift in the compliance–performance frontier:** Capabilities previously associated with mixed-license corpora are reachable with a permissive-first, risk-mitigated approach
- **Front-loading instruction & reasoning data into pretraining** is more token-efficient than relying on post-training alone, 36x fewer tokens than SmolLM2-Instruct
- **Decontamination verified:** Performance holds on decontaminated benchmarks and with contaminated shards removed
- **Scalable recipe:** Path to multi-trillion token regime via subset upsampling, multilingual expansion, and synthetic growth

Code: https://github.com/ontocord/mixturevitae
Dataset: https://huggingface.co/datasets/ontocord/MixtureVitae-v1
Paper: https://arxiv.org/abs/2509.25531